

Estimating Dewpoint Temperature for Peru Potato Planting Locations Using an Anomaly Local Linear Regression Model

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What is this project?

‘tdew_estimation’ is a scientific software tool that estimates **daily dewpoint temperature** (often called “Td” or “td”) in places or periods where that measurement is missing. Dewpoint temperature is a key climate variable describing the temperature at which air becomes saturated with moisture and dew can form. This tool was built to support **Simcast workflows** (for example, in agriculture or environmental science), where having dewpoint data is essential but direct measurements may be unavailable.

What is it for?

Many scientific applications—like forecasting plant disease risks, running environmental models, or filling out weather records—need complete, accurate records of daily dewpoint temperature for different locations (called “grid cells”). However, not all weather stations measure dewpoint every day; sometimes, important days or locations are missing this measurement.

This tool helps by **reconstructing or estimating** the missing values using advanced statistical techniques. Specifically, it uses:

All of this is done at **high spatial resolution** (many grid cells) and for cases where dewpoint is missing or has gaps.

What is Dask, and why do we need it for this?

Calculating all the required estimates for dewpoint across thousands of locations and all days in a year is a huge job—much too large for even a powerful personal laptop. This means:

- The process involves millions of mini-calculations, since every location (sometimes tens of thousands) and every day of the year needs its own custom equation and data check.
- Doing this work one-by-one would take weeks even on a powerful computer, and it might run out of memory or crash.

Dask is a technology that lets us divide this giant task into smaller pieces and spread the pieces across many computers working together. Instead of one computer doing everything, Dask acts like an efficient manager in a large office—giving each worker a small part of the job. Everyone works in parallel, then results are combined at the end.

This project is designed to use clusters of computers—sometimes with dozens of machines—to finish in hours instead of weeks.

What does this mean for users?

- You don’t need to be a coding expert, but the computation won’t run on a normal laptop or desktop alone. It needs access to a computing cluster (“cloud” or “HPC” resources), set up by data managers.
- Dask ensures the process is robust: if one “worker” fails or needs a break, the others continue, and the task gets finished.
- This is similar to how weather agencies run large climate models.

Summary: Using Dask makes massive, complex scientific estimation possible and practical, so you can have high-quality dewpoint data for modeling or decision making—even across entire countries—without waiting forever. ## What’s the science/math behind it?

- **Climatology:** For each location and each day of the year, the tool computes the “average” dewpoint temperature (and minimum temperature) based on available historic data. This is called the daily “climatology.”
- **Anomalies:** On any day, the difference between the observed (or missing) value and this “typical” value is called the “anomaly.” The tool models these anomalies instead of the raw values.
- **Local Linear Regression Model:** For each location and each time of year, it predicts dewpoint anomalies using:
 - The anomaly in minimum temperature for that day
 - Recent values (from previous days) of both dewpoint and temperature anomalies
 - A statistical technique called “weighted local linear regression” that gives more influence to data from nearby dates on the calendar (so predictions in July rely mostly on past Julys).

Math in Simple Terms

The tool fits an equation like this for each location and each time of year:

$$TD_{\text{anom}} \approx \text{constant} + (\text{TMIN}_{\text{anom}} \times \text{coeff}_1) + (\text{yesterday's } TD_{\text{anom}} \times \text{coeff}_2) + \dots$$

- “TD_anom” is the anomaly (difference from normal) in dewpoint.
- “TMIN_anom” is the anomaly in minimum temperature.
- There are also terms using values from one and two days earlier (lags).

By fitting these equations, the tool learns how dewpoint anomalies depend on temperature anomalies and their recent history, separately for each location and time of year.

- **Regression weights** are highest for days closest to the day being predicted, using a “tricube” kernel—which just means points further away on the calendar matter less.

What is the expected result or output if I use this? What files do I get?

If you use this tool, you can expect the following outputs:

1. **Climatology table (`daily_climatology.parquet`)**
For each location and each day of the year: the typical (mean) dewpoint and minimum temperature.
2. **Coefficients table (`llr_coeffs_anomaly_final_direct.parquet`)**
For each location and each day of the year: the fitted regression coefficients. These are the numbers that tell you, mathematically, how dewpoint is expected to vary as minimum temperature changes, and how yesterday’s/more recent conditions affect today.
3. **(Optional/Intermediate) Chunk files**
Temporary files created when doing parallel calculation for faster or distributed processing.
4. **(If needed) Patch files**
If the first attempt leaves some days incomplete (missing some combinations of location/day), repair files are created for just those gaps and then merged in.

In practical terms:

- After running this tool, you have all the necessary ingredients to reconstruct a full daily dewpoint record for your area and time period—either filling in missing days or forecasting new ones—by applying these coefficients to available minimum temperature data.

Who should use this?

- Scientists, agricultural experts, and environmental analysts who need complete dewpoint time series for modeling, risk assessment, or gap-filling.